In [2]:
```
import numpy as np     #importing numpy and pandas
import pandas as pd
```

In [3]:
```
ds = pd.read_excel("C:\\Users\\Tanishq Solanki\\Downloads\\customer_shopping_dat
ds
```

Out[3]:

	invoice_no	customer_id	gender	age	category	quantity	price	payment_r
0	I138884	C241288	Female	28	Clothing	5	1500.40	Cre
1	I317333	C111565	Male	21	Shoes	3	1800.51	De
2	I127801	C266599	Male	20	Clothing	1	300.08	
3	I173702	C988172	Female	66	Shoes	5	3000.85	Cre
4	I337046	C189076	Female	53	Books	4	60.60	
...	...	...	...	...	...	...	...	
99452	I219422	C441542	Female	45	Souvenir	5	58.65	Cre
99453	I325143	C569580	Male	27	Food & Beverage	2	10.46	
99454	I824010	C103292	Male	63	Food & Beverage	2	10.46	De
99455	I702964	C800631	Male	56	Technology	4	4200.00	
99456	I232867	C273973	Female	36	Souvenir	3	35.19	Cre

99457 rows × 10 columns

In [4]:
```
ds.info()  #basic information of data
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99457 entries, 0 to 99456
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   invoice_no      99457 non-null  object
 1   customer_id     99457 non-null  object
 2   gender          99457 non-null  object
 3   age             99457 non-null  int64
 4   category        99457 non-null  object
 5   quantity        99457 non-null  int64
 6   price           99457 non-null  float64
 7   payment_method  99457 non-null  object
 8   invoice_date    99457 non-null  datetime64[ns]
 9   shopping_mall   99457 non-null  object
dtypes: datetime64[ns](1), float64(1), int64(2), object(6)
memory usage: 7.6+ MB
```

In [5]:
```
ds.describe()    #basic statistics from data
```

Out[5]:

	age	quantity	price	invoice_date
count	99457.000000	99457.000000	99457.000000	99457
mean	43.427089	3.003429	689.256321	2022-02-04 02:46:59.783424
min	18.000000	1.000000	5.230000	2021-01-01 00:00:00
25%	30.000000	2.000000	45.450000	2021-07-19 00:00:00
50%	43.000000	3.000000	203.300000	2022-02-05 00:00:00
75%	56.000000	4.000000	1200.320000	2022-08-22 00:00:00
max	69.000000	5.000000	5250.000000	2023-03-08 00:00:00
std	14.990054	1.413025	941.184567	NaN

In [6]:
```
ds.isnull().sum()    #checking the null values
```

Out[6]:
```
invoice_no        0
customer_id       0
gender            0
age               0
category          0
quantity          0
price             0
payment_method    0
invoice_date      0
shopping_mall     0
dtype: int64
```

In [7]:
```
ds.shape   #no. of rows and no. of columns
```

Out[7]:
```
(99457, 10)
```

In [8]:
```
ds
```

Out[8]:

	invoice_no	customer_id	gender	age	category	quantity	price	payment_r
0	I138884	C241288	Female	28	Clothing	5	1500.40	Cre
1	I317333	C111565	Male	21	Shoes	3	1800.51	De
2	I127801	C266599	Male	20	Clothing	1	300.08	
3	I173702	C988172	Female	66	Shoes	5	3000.85	Cre
4	I337046	C189076	Female	53	Books	4	60.60	
...	...	...	...	...	...	...	...	
99452	I219422	C441542	Female	45	Souvenir	5	58.65	Cre
99453	I325143	C569580	Male	27	Food & Beverage	2	10.46	
99454	I824010	C103292	Male	63	Food & Beverage	2	10.46	De
99455	I702964	C800631	Male	56	Technology	4	4200.00	
99456	I232867	C273973	Female	36	Souvenir	3	35.19	Cre

99457 rows × 10 columns

In [9]:
```
ds["price"].sum()    # Total Sales
```

Out[9]:
```
np.float64(68551365.91)
```

In [10]:
```
ds.groupby("category")["price"].sum().sort_values(ascending=False)    #category
```

Out[10]:
```
category
Clothing           31075684.64
Shoes              18135336.89
Technology         15772050.00
Cosmetics           1848606.90
Toys                1086704.64
Food & Beverage      231568.71
Books                226977.30
Souvenir             174436.83
Name: price, dtype: float64
```

In [11]:
```
ds.groupby("shopping_mall")["price"].sum().sort_values(ascending=False)     #sho
```

Out[11]:
```
shopping_mall
Mall of Istanbul     13851737.62
Kanyon               13710755.24
Metrocity            10249980.07
Metropol AVM          6937992.99
Istinye Park          6717077.54
Zorlu Center          3509649.02
Cevahir AVM           3433671.84
Viaport Outlet        3414019.46
Emaar Square Mall     3390408.31
Forum Istanbul        3336073.82
Name: price, dtype: float64
```

In [12]:
```
ds.groupby("gender")["price"].sum().sort_values(ascending=False)    #distrubutio
```

Out[12]:
```
gender
Female    40931801.62
Male      27619564.29
Name: price, dtype: float64
```

In [13]:
```
bins =[18,25,35,50,70]                    # created a two list and then used pd.c
labels = ["young","middle age","adult","senior"]
ds['age_group'] = pd.cut(ds['age'], bins=bins, labels=labels)
```

In [14]:
```
ds.groupby("age_group")["price"].sum().sort_values(ascending=False)     #age_gro
```

```
C:\Users\Tanishq Solanki\AppData\Local\Temp\ipykernel_2212\2333284885.py:1: Futur
eWarning: The default of observed=False is deprecated and will be changed to True
in a future version of pandas. Pass observed=False to retain current behavior or
observed=True to adopt the future default and silence this warning.
  ds.groupby("age_group")["price"].sum().sort_values(ascending=False)     #age_gr
oup wise distribution
```

Out[14]:
```
age_group
senior        24833266.32
adult         20171970.76
middle age    13101356.40
young          9233308.81
Name: price, dtype: float64
```

In [15]:
```
ds.groupby("payment_method")["price"].sum().sort_values(ascending=False)    #pay
```

Out[15]:
```
payment_method
Cash           30705030.98
Credit Card    24051476.93
Debit Card     13794858.00
Name: price, dtype: float64
```

In [16]:
```
ds["ord_month"]= ds["invoice_date"].dt.month  #added a column for months
ds
```

Out[16]:

	invoice_no	customer_id	gender	age	category	quantity	price	payment_r
0	I138884	C241288	Female	28	Clothing	5	1500.40	Cre
1	I317333	C111565	Male	21	Shoes	3	1800.51	De
2	I127801	C266599	Male	20	Clothing	1	300.08	
3	I173702	C988172	Female	66	Shoes	5	3000.85	Cre
4	I337046	C189076	Female	53	Books	4	60.60	
...	...	...	...	...	...	...	...	
99452	I219422	C441542	Female	45	Souvenir	5	58.65	Cre
99453	I325143	C569580	Male	27	Food & Beverage	2	10.46	
99454	I824010	C103292	Male	63	Food & Beverage	2	10.46	De
99455	I702964	C800631	Male	56	Technology	4	4200.00	
99456	I232867	C273973	Female	36	Souvenir	3	35.19	Cre

99457 rows × 12 columns

In [17]:
```
ds.groupby("ord_month")["price"].sum()      #month wise analysis
```

Out[17]:
```
ord_month
1     7932626.63
2     7235297.32
3     6007346.21
4     5102478.76
5     5346926.82
6     5160345.74
7     5552023.57
8     5270542.03
9     5115214.27
10    5538258.09
11    4995141.11
12    5295165.36
Name: price, dtype: float64
```

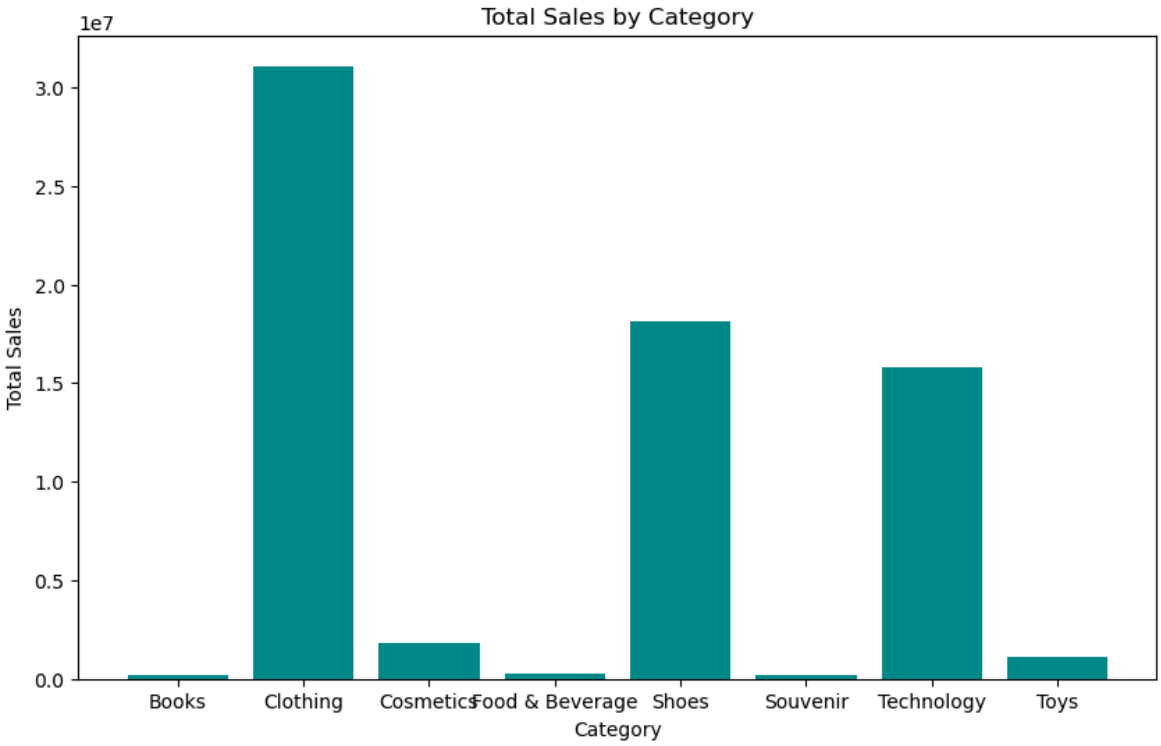
In [18]:
```
ds.groupby("category")["quantity"].sum().sort_values(ascending=False)    #most qu
```

Out[18]:
```
category
Clothing           103558
Cosmetics           45465
Food & Beverage     44277
Toys                30321
Shoes               30217
Technology          15021
Books               14982
Souvenir            14871
Name: quantity, dtype: int64
```

In [19]:
```
import matplotlib.pyplot as plt     # importing matplotlib for visualisatuion
```

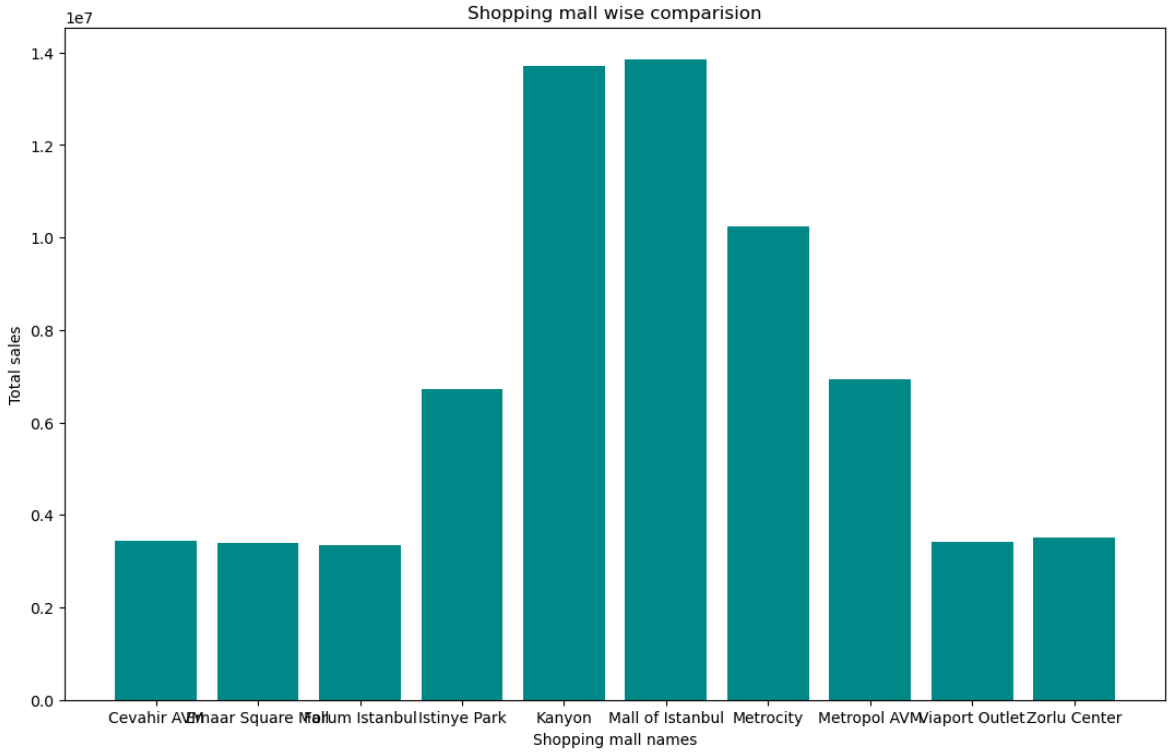
In [22]:
```
category_sales = ds.groupby("category")["price"].sum()   #Have to make calculati
```

In [70]:
```
plt.figure(figsize=(10,6))                    #plotting bar graph using index ana

plt.bar(category_sales.index, category_sales.values, color='#048787')

plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.title("Total Sales by Category")

plt.show()
```



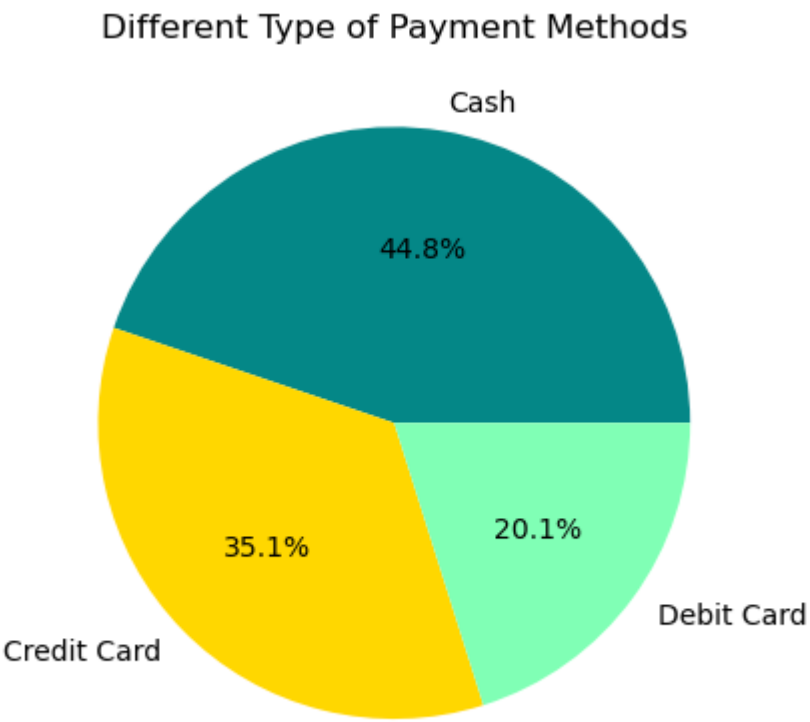
In [24]:
```
mall_sales =ds.groupby("shopping_mall")["price"].sum()    #making group by for a
```

In [69]:
```
plt.figure(figsize=(13,8))                                #making bar graph usi
plt.bar(mall_sales.index,mall_sales.values,color='#048787')
plt.xlabel("Shopping mall names")
plt.ylabel("Total sales")
plt.title("Shopping mall wise comparision")
plt.show()
```



In [36]:
```
payment_type= ds.groupby("payment_method")["price"].sum() #using group by for ma
```

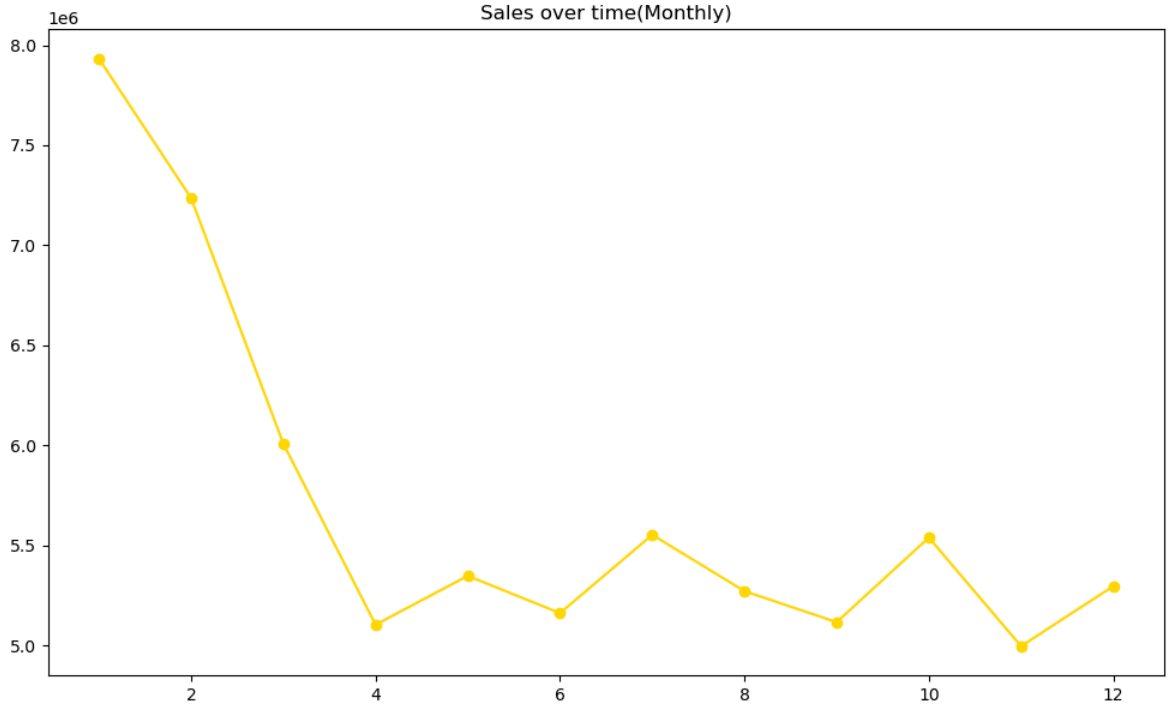
In [68]:
```
colors = ["#048787", "gold","#80FFB5"]

plt.pie(payment_type.values,labels=payment_type.index,autopct="%1.1f%%",colors=c
plt.title("Different Type of Payment Methods")
plt.show()
```



In [ ]:

In [45]:
```
monthly_sale = ds.groupby("ord_month")["price"].sum()
```

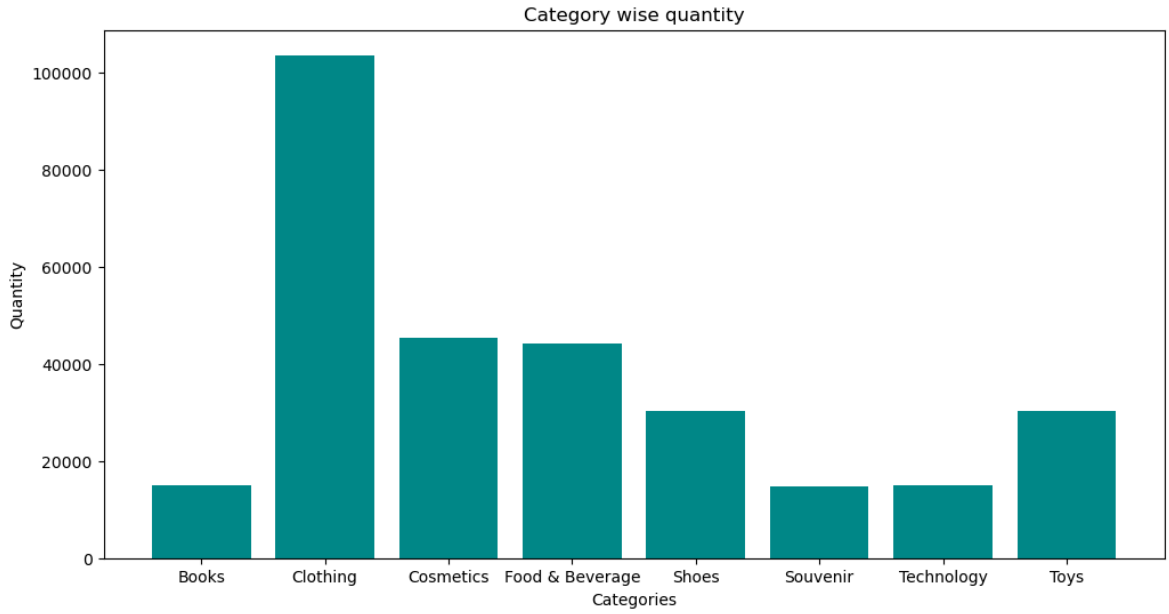
In [63]:
```
plt.figure(figsize=(12,7))
plt.plot(monthly_sale.index,monthly_sale.values,marker='o',color='gold')
plt.title("Sales over time(Monthly)")
plt.show()
```



In [ ]:

In [48]:
```
qnt = ds.groupby("category")["quantity"].sum()
```

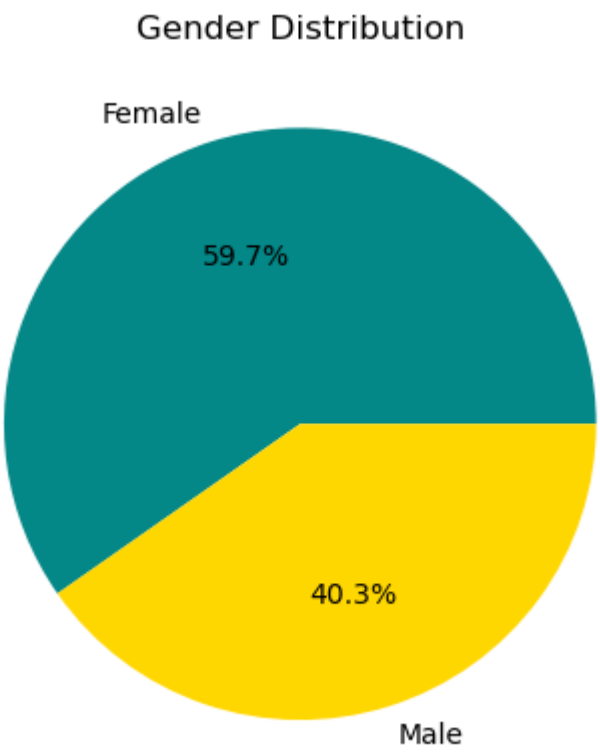
In [62]:
```
plt.figure(figsize=(12,6))
plt.bar(qnt.index,qnt.values,color='#048787')
plt.xlabel("Categories")
plt.ylabel("Quantity")
plt.title("Category wise quantity")
plt.show()
```



In [ ]:

In [51]:
```
gender_wise = ds.groupby("gender")["price"].sum()
```

In [61]:
```
colors = ["#048787", "gold"]
plt.pie(gender_wise.values,labels=gender_wise.index,autopct="%1.1f%%",colors=col
plt.title("Gender Distribution")
plt.show()
```



In [ ]:

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